

BUSINESS INTELLIGENCE REPORT: BIRD SPECIES OBSERVATION ANALYSIS

Executive Summary

This report synthesizes the findings from the **Bird Species Observation Analysis** project, which evaluates avian biodiversity across forest and grassland ecosystems. Utilizing a dataset of over **14,264 observations**, the project employs a full-stack data pipeline—from Python-based preprocessing and SQL warehousing to Power BI visualization—to provide actionable insights for wildlife conservation and land management.

High-Level Metrics

14,264	126	25	7:00 AM
Total Observations	Unique Species Tracked	At-Risk Species	Peak Activity Hour

Core Problem Solved

"Raw observational data is often fragmented and lacks environmental context. This project builds a structured analytical framework to quantify how habitat types (Forest vs. Grassland) and climatic factors influence bird density and diversity, enabling data-driven conservation strategies."

Key Findings & Insights

1. Habitat Diversity & Preference

- **Grassland Dominance:** The **MONO (Grassland)** unit emerged as the most diverse, hosting **93 unique species**.
- **Northern Cardinal Prevalence:** This species remains the most frequently observed, indicating high adaptability across both studied ecosystems.
- **Flyover Trends:** Approximately **8.75%** of observations were flyovers (birds in transit), which helps distinguish between resident populations and migratory transit patterns.

2. Temporal & Environmental Patterns

- **The Morning Peak:** Avian activity is highest at **07:00**, coinciding with the "dawn chorus." Data suggests a sharp decline in observable activity after 10:00 AM.
- **Seasonal Surge:** **June** is the peak month for surveys and sightings, likely tied to breeding and nesting cycles in the Spring/Summer transition.
- **Weather Impact:** Most sightings occurred during "Partly Cloudy" conditions with "Light Air" movement, suggesting that extreme weather significantly suppresses detection rates.

3. Conservation Indicators

- **Watchlist Status:** **25 species** were identified as "At-Risk" based on PIF Watchlist statuses. These species require prioritized habitat protection.
- **At-Risk Density:** A correlation exists between specific forest plots (e.g., ANTI-0036) and the presence of regional stewardship species, marking these as high-value conservation zones.

Operational Impact & Pipeline Architecture

Challenge	Technical Solution	Outcome
Data Inconsistency	Python (Pandas/NumPy) preprocessing for missing values and outlier detection.	A clean, 100% validated dataset ready for modeling.
Siloed Observations	Star Schema SQL Warehouse (dim_species, dim_plot, fact_observations).	Enabled complex YoY growth and running total queries.
Manual Reporting	Power BI Interactive Dashboard with DAX measures for "At-Risk" counts.	Real-time filtering by habitat, season, and species status.

Complex Logic	SQL CTEs for temporal trends and species diversity indexing.	Decoupled business logic from the UI for faster report rendering.
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Strategic Recommendations

- **Focus on 'MONO' Units:** Given the high species richness in Grasslands, conservation funding should prioritize these areas to protect the largest variety of species within a single administrative unit.
- **Standardize Survey Windows:** To optimize resource allocation, field observations should be concentrated between **06:00 and 09:00**, as data shows a significant "observation-per-hour" drop-off thereafter.
- **Expand 'At-Risk' Monitoring:** Phase 2 should include a "Species Stability Index" to track if the 25 at-risk species are increasing or decreasing in sighting frequency year-over-year.

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Project: Bird Species Observation Analysis

Tools used: Python (EDA), SQL (Storage/Transformation), Power BI (Visualization)